Lab: Switch CLI IOS Basics

CNSA166: Internetworking Devices

Use your initial Packet Tracer file for this lab (the first Packet Tracer lab of this module). This lab will be time consuming; make sure you take your time and do it right. We will continue to build off of this. Before you do this lab, make sure you have static routing set up and can ping between networks. This should be a nicely documented lab with all questions answered. Your lab documentation and work are graded strictly moving forward for clarity, accuracy, and connectivity. Take these labs seriously, as the final exam will be heavily built off of what you are doing. Questions, ask.

Refer to chapters 4, 5, and 6 in your textbook for command assistance.

NOTE: You're going to repeat many of these steps on all six switches in your PT network. Consider working through everything on one switch (familiarize yourself on the first run), then go back and repeat all steps on the other switches. This might help alleviate the bouncing back and forth between switch CLI interfaces.

1. You should have six switches in your PT network – Switch0 through Switch5. Using the CLI interface tab name each switch with the following format: AdminSW0, AdminSW1, AdminSW2, etc.

2. What command did you use to name the switches?
   I had to first use 'en' to get into privileged exec mode, then 'configure terminal' to get into global configuration mode, then use 'hostname AdminSW0'
3. Secure privileged EXEC mode on all switches with the password ***cisco1.***
   Use 'en' to get into privileged exec mode, then 'configure terminal' to get into global configuration mode, then 'enable secret cisco1' to set the password
4. Secure the console port on all switches by configuring a local login for username ***bob*** with password ***cisco.***
   Open the CLI menu, 'en' to go into privileged exec, 'configure terminal' to get into global configuration, type 'line console 0' to get into line mode, type 'username bob password cisco' then type 'login local' while in line mode still. Type 'end' to go back to privileged exec mode, then 'configure terminal' to get into global config. Put

‘service password-encryption’ and then exit all the way out and back in to test that it’s working

5. Exit the CLI switch interface and log back in to test your passwords. Did the console ask you for a username/password pair?
   Yes
6. Secure the first five telnet lines on all switches with the password ***cisco2*** that should be asked for when someone attempts to log in via a telnet connection.
   Open CLI menu, ‘en’ to get into privileged exec, ‘configure terminal’ to get into global config, ‘line vty 0 4’ to get into line mode and the part you need, ‘password cisco2’ to set the password, ‘login’ to set the password to actually be used, ‘transport input telnet’ to ensure security, ‘service password-encryption’ to encrypt passwords, ‘end’ to leave
7. How do you check if your telnet passwords are active?
   Get into privileged exec mode and then type ‘show run’ The running configuration will pop up and you have to press enter or space to make more lines pop up until your get to where it says ‘line vty 0 4’ and check that it’s showing a password under that setting.
8. In order to do step 7, connecting via telnet, you need to have an IP address assigned to your switch. How do you do this?
   ‘en’ to get into privileged exec, ‘configure terminal’ to go into global config, ‘interface vlan 1’ to enter VLAN config mode, ‘ip address 172.16.1.9 255.255.255.0’ to set the ip address and subnet mask, ‘no shutdown’ to keep the interface running.
9. Go ahead and set an IP address on all six switches.

10. Also configure the default gateway for each switch. How do you configure this? From what mode?
    Go into configure terminal mode then ‘ip default-gateway 172.16.1.1’
11. You might want to consider turning off name resolution if you make a lot of typos. Do this on all six switches. What command did you use?
    ‘en’ to get into privileged exec mode, ‘configure terminal’ to go into global config, no ip domain-lookup, to disable name resolution
12. Before you get too far into your configurations, you should save the work you have done so far. Note – if you restart a router or switch in the simulator and you haven’t saved your configuration file, it will disappear. Just like Word docs, save often. What command do you use to save your running configuration into NVRAM?
    Get into privileged exec mode with ‘en’ then type ‘write’
13. It’s also a good idea to use the ***logging synchronous*** command on the console and telnet lines. Why? What does this command do?

It prevents unsolicited log messages and debug output from interrupting what you're typing at the moment

14. How many commands does the switch store in its history by default?

    16

15. What command would you use to change the number of commands held in the history?

    'Terminal history size' in privileged exec mode

16. Does the switch retain the command history if you exit all the way out and re-enter through the console connection?

    No

17. The Cisco routers and switches will log out a console or telnet line after a period of inactivity. What is the default timeout?

    10 minutes

18. How would you set the console timeout to six minutes? What mode are you in to configure this?

    'en' to go to privelged exec, 'configure terminal' to go to global config, 'console timeout 6'

19. Explore the debug commands available on one of your switches. What can you 'debug' on your switch? Why do we use debug commands?

    Vlan, stp, ip, ip information. To fix small issues in a switch without resetting unrelated things

20. Challenge: configure AdminSW0 to accept SSH logins. Document your process here: What commands did you use, in what mode, and why? How did you test it?

    'en' to go the privileged exec mode, ' configure terminal' to go to global config 'ip domain-name connect.com' to make the ip domain name 'crypto key generate rsa' to make the keys and then choose a number of bits '1024' 'username bob password cisco' to make an account that can sign into it 'ip ssh version 2' for better security 'line vty 0 15' 'transport input ssh' 'login local' so that it will accept ssh connections.

    I tested it by going into the command prompt of a connected PC and typing in 'ssh -1 bob 172.16.1.9' then typing in the password.